

VIRGINIA COMMONWEALTH UNIVERSITY
Department of Mathematics & Applied Mathematics
MATH 756 – Topics in Graph Theory: Independent Sets
Fall 2019

Instructor: Dr Larson

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Office Hours: 11:00-12:00 MWF

Prerequisite(s): MATH 656.

Text and Other Materials: None required.

Topics and Goal: A detailed study of the theory of independent sets in graphs, bounds for the independence number, and algorithms for finding maximum independent sets and the independence number.

Attendance: There is no attendance policy per se, but there will be homework and in-class assignments that are due.

Expectations:

- You are expected to attend class, complete homework, and ask questions during class or office hours.
- When presenting your work, I expect you to show all significant steps that are used to complete each problem. In cases where work is missing, you will not be given full credit.
- I encourage you to work with others on homework problems, however, any assignments to be turned in must be written up on your own. If you work with others, you must write who you worked with on your assignment.
- Please write neatly on all assignments to be graded; exceptionally messy work may not be graded.
- Only selected homework problems will be graded; other problems will be graded for completion.
- **There are no make-ups on in-class assignments.** I will drop your three lowest in-class assignments, assuming that you couldn't come to class for excusable reasons.
- Make up tests will be considered under exceptional circumstances: if you miss a test and want to be considered for a make-up, you *must* contact me immediately.

Tests and Determination of Grades:

There will be two equally weighted tests. Here is the *tentative* schedule:

Test #1, Mon., Oct. 14

Test #2, Fri., Dec. 13, 8:00-10:50 (during our scheduled **FINAL** time).

- The tests are closed-book and closed-notes.
- The tests will be *closely based on* the in-class assignments and assigned homework.
- Use of calculators or other computing technology is not allowed on the tests.
- Tests are written under the assumption that you are studying the material at least 6 hours per week outside of class.

Your final average will be computed as follows:

Tests:	50% (25% each)
Homework:	20%
In-class assignments:	20%
Project	10%

Grade Scale: The 10-point scale: 90-100 A, 80-89 B, etc.

Important Dates to Know:

- Reading Day: Fri., Oct. 18
- Classes end: Mon., Dec. 9

VCU Syllabus Statement: Students should visit <http://go.vcu.edu/syllabus> and review **all** syllabus statement information. The full university syllabus statement includes information on safety, registration, the VCU Honor Code, student conduct, withdrawal and more.